

Chapter 1 - The Internet

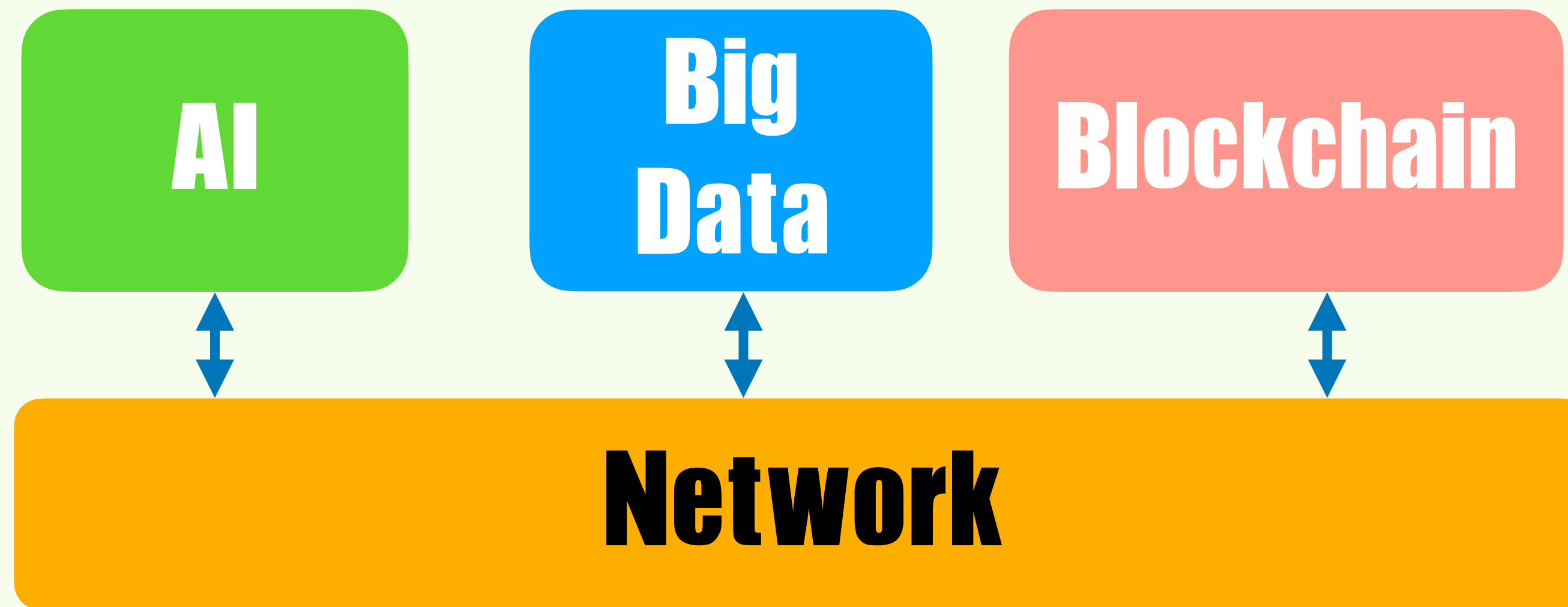
The tech that connect civilisations

Computer Network

Why is Computer Network Important?

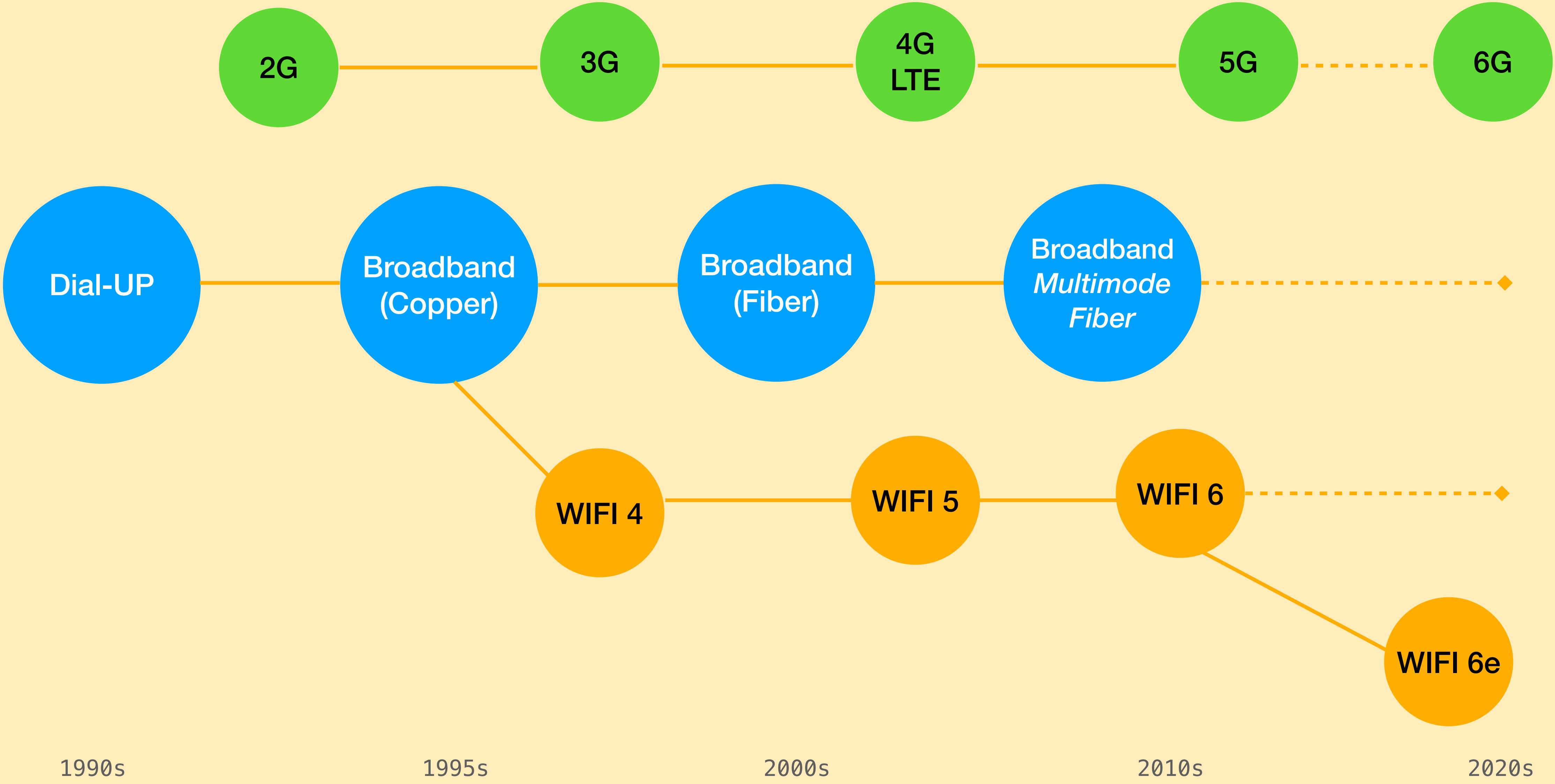


Why is Computer Network Important?



Tech Paradigm (2023)

Timeline



Networking Technology

Networking Technology

A **wired** network uses cables to connect devices to the network, while a **wireless** network uses radio waves to connect devices to the network. Wireless networks are more convenient as they do not require physical connections, but wired networks may provide faster speeds and more reliable connections.

Wired

- Ethernet, Fast Ethernet, Gigabit Ethernet
- Serial
- Console
- Aux

Wireless

- Bluetooth
- Cellular - 3G, 4G, 5G
- Near Field Communication (NFC)
- WiFi Direct

WIRED vs WIFI

Wired vs Wireless

Speed:

wired networks tend to be faster than wireless networks. This is because wired connections, such as Ethernet, have a dedicated and direct connection between the devices, while wireless connections, such as Wi-Fi, have to share a limited amount of spectrum and may experience interference from other devices.

Convenience:

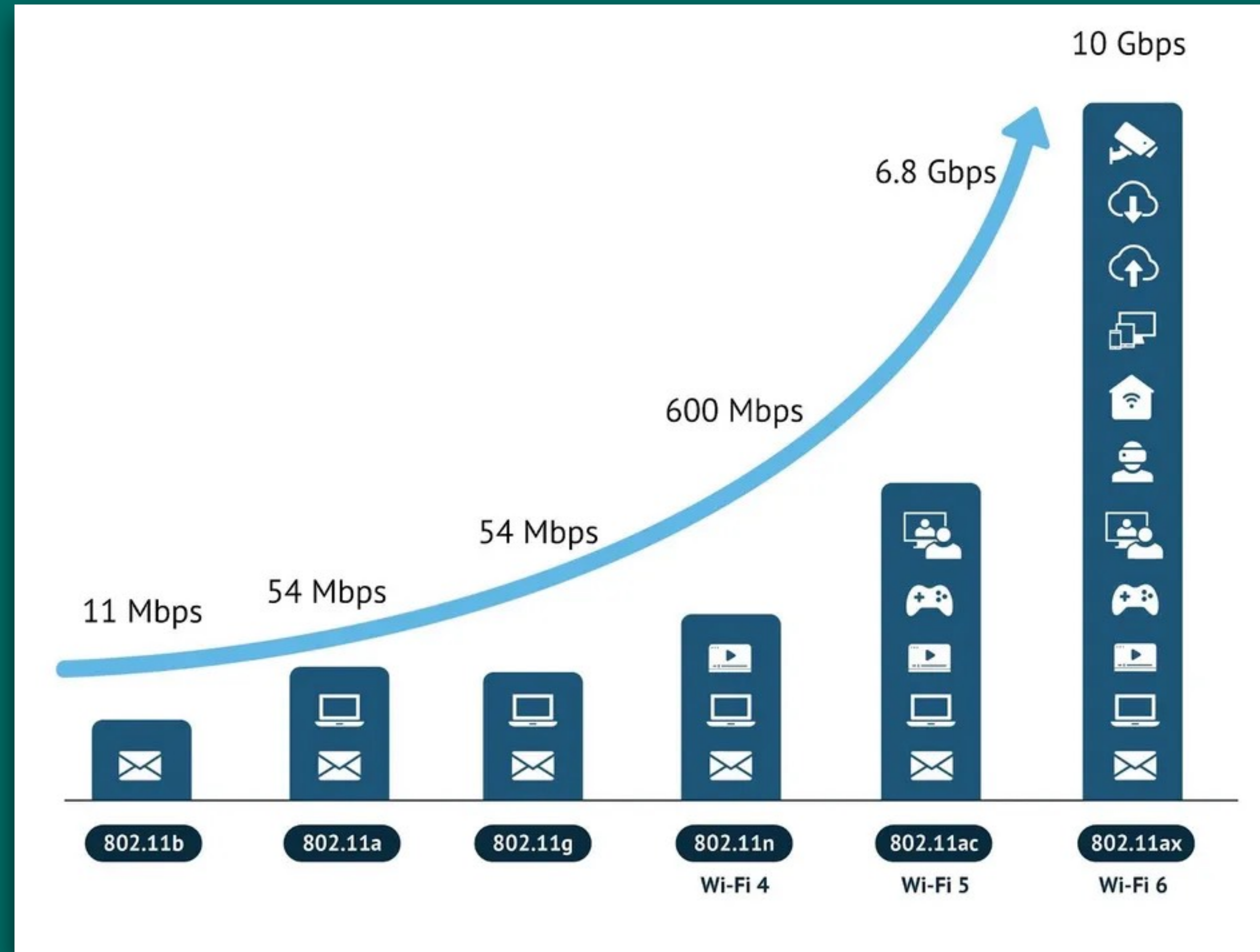
Wireless connection is more convenient as it does not require physical cables, but it may be slower and less reliable than a wired connection

WIFI vs 5G

WiFi Family Photo

Year	WiFi Versions/Protocols	Old name	New name
1999	First Generation	WiFi 802.11b	WiFi 1
1999	Second Generation	WiFi 802.11a	WiFi 2
2003	Third Generation	WiFi 802.11g	WiFi 3
2009	Four Generation	WiFi 802.11n	WiFi 4
2014	Fifth Generation	WiFi 802.11ac	WiFi 5
2019	Sixth Generation	WiFi 802.11ax	WiFi 6

WIFI versions



Wireless vs 5G

Wi-Fi 6 and 5G are wireless technologies that connect devices to a network, but they operate at different frequencies and have different capabilities.

Wi-Fi 6

- ◆uses the 2.4 GHz and 5 GHz bands
- ◆has a maximum theoretical speed of 9.6 Gbps
- ◆increased capacity, lower latency, and better energy efficiency.

5G

- ◆is a cellular network technology that is designed for mobile devices
- ◆5G uses the sub-6GHz and millimeter wave
- ◆maximum theoretical speed of 20 Gbps.
- ◆support for low-latency and high-bandwidth applications like VR and AR.

Network Metrics

Network Metrics

Bandwidth (Mbps/Gbps)

The maximum amount of data that can be transferred in a given amount of time

Throughput

The maximum amount of data that is actually transferred in a given amount of time

Latency (delay)

The time taken for a data to go from source host to the destination host and back to the source host

Bandwidth (Mbps/Gbps)

is the amount of data that can be transmitted over a network at a given time.

For example, if you subscribed to Unifi 100Mbps, you can get up to 100Mbps of connection speed.

In a network with a bandwidth of 100 Mbps. If there are multiple devices connected to the network. The 100Mbps bandwidth is shared among all the devices transmitting data simultaneously. Thus, the bandwidth of each device is less than 100Mbps.

Throughput

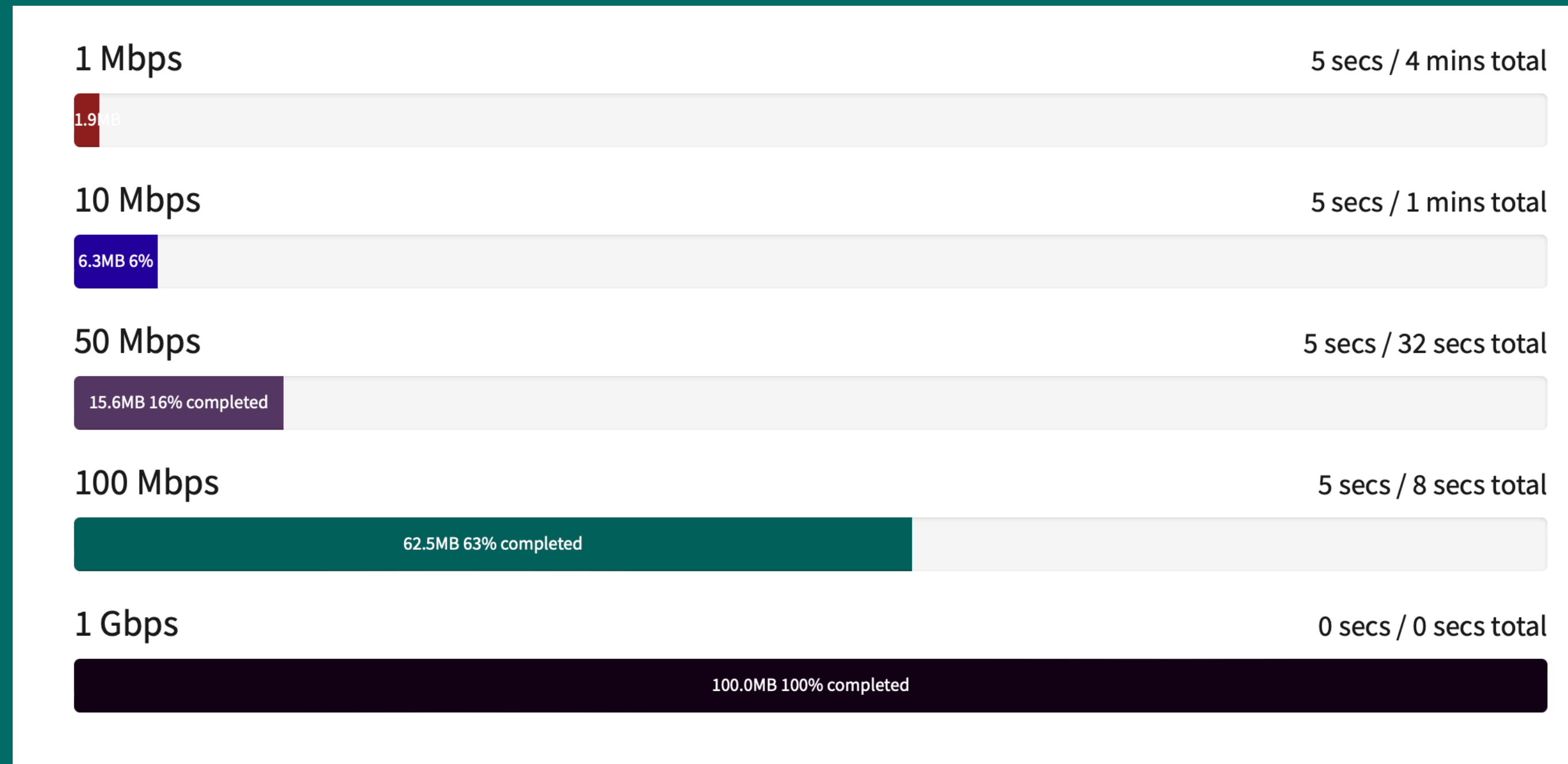
The maximum amount of data that is **actually** transferred in a given amount of time

The throughput of a network can be affected by various factors such as network congestion, signal interference, and protocol overhead.

Latency (delay)

The time taken for a data to go from source host to the destination host and back to the source host

How fast is fast?



Wired:
Google Fiber (2000Mbps)

Wireless:
Verizon millimetre wave 5G (2000Mbps)

Why your Internet speed is slower than advertised?

Factors that determine your Internet Speed

Cable Types

- Cat5, 5e and 6 provides max 1000Mbps transmission speed
- Cat 6a, 7 provides max 10,000Mbps transmission speed

Category	Max speed transmission	Max bandwidth
Cat 5	100 Mbps	100 MHz
Cat 5e	1,000 Mbps	100 MHz
Cat 6	1,000 Mbps	250 MHz
Cat 6a	10,000 Mbps	500 MHz
Cat 7	10,000 Mbps	600 MHz
Cat 7a	10,000 Mbps	1,000 MHz

Network Devices Interface

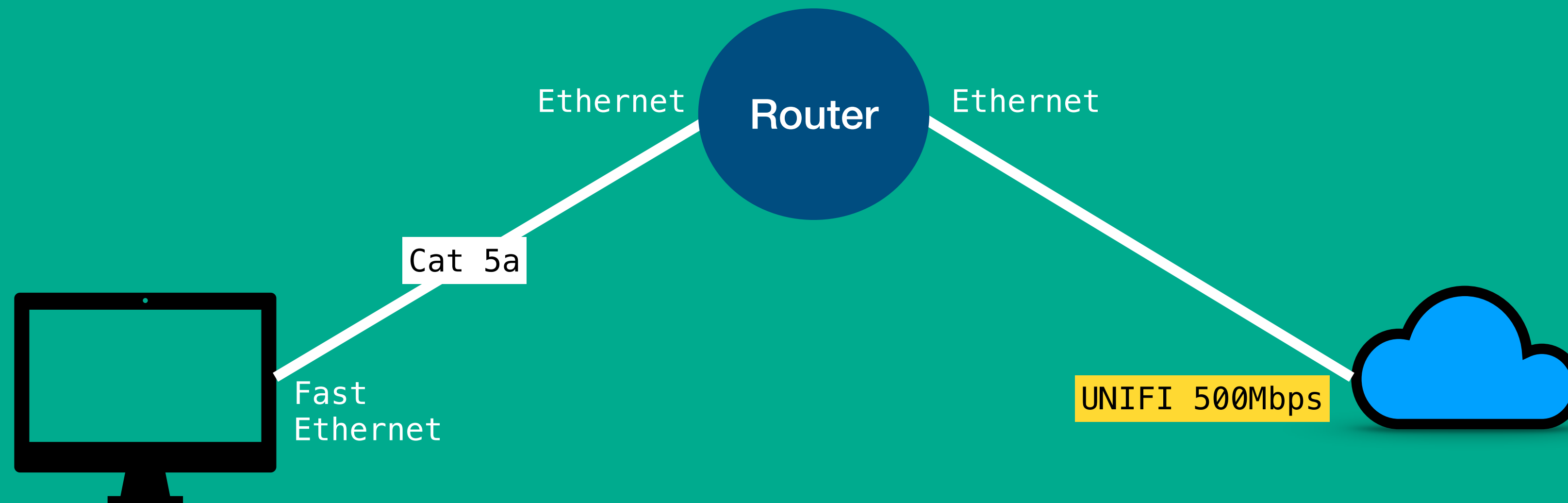
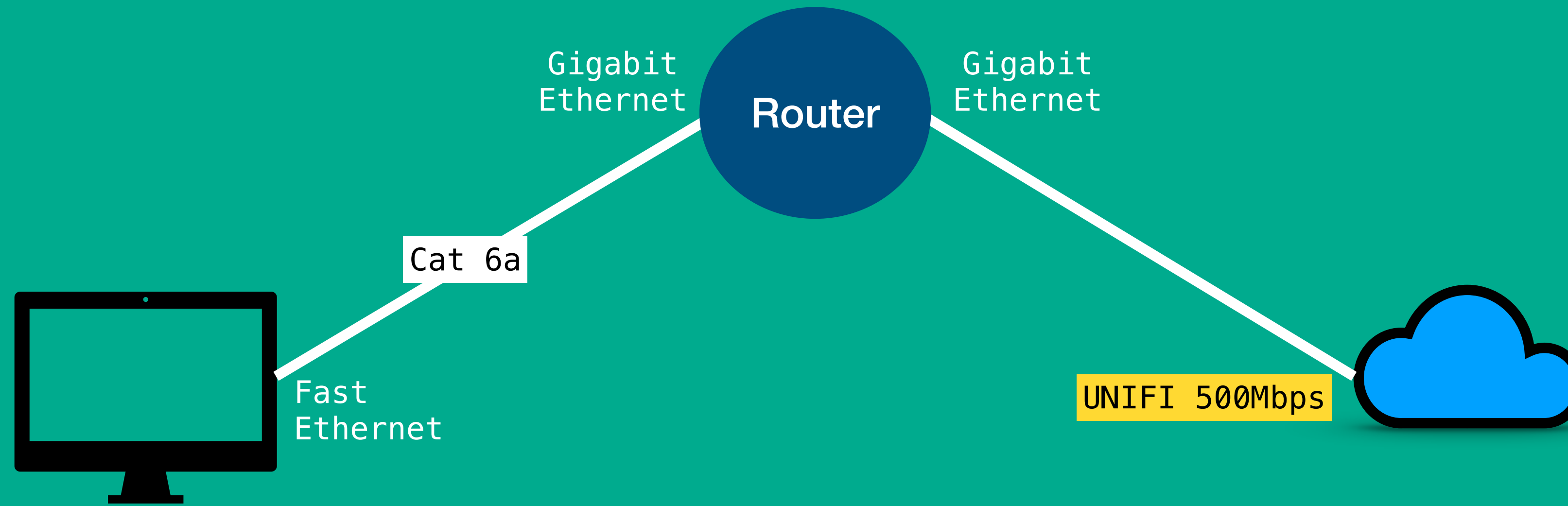
- Ethernet - 10Mbit/s
- Fast Ethernet - 100Mbit/s
- Gigabit Ethernet - 1000Mbit/s

Interferences/Attenuation

- Wireless signal degrades in a crowded area of electrical appliances
- Wired signal deteriorate when travel over long distance

ISP Promises

- ISP speed are advertised as best efforts, only 80% of link speed guarantee
- ISP speed are subject to fair usage policy, more crowds, slower speed



The Internet

Google Fiber
AWS Network
Azure
Facebook
Microsoft

Google Fiber

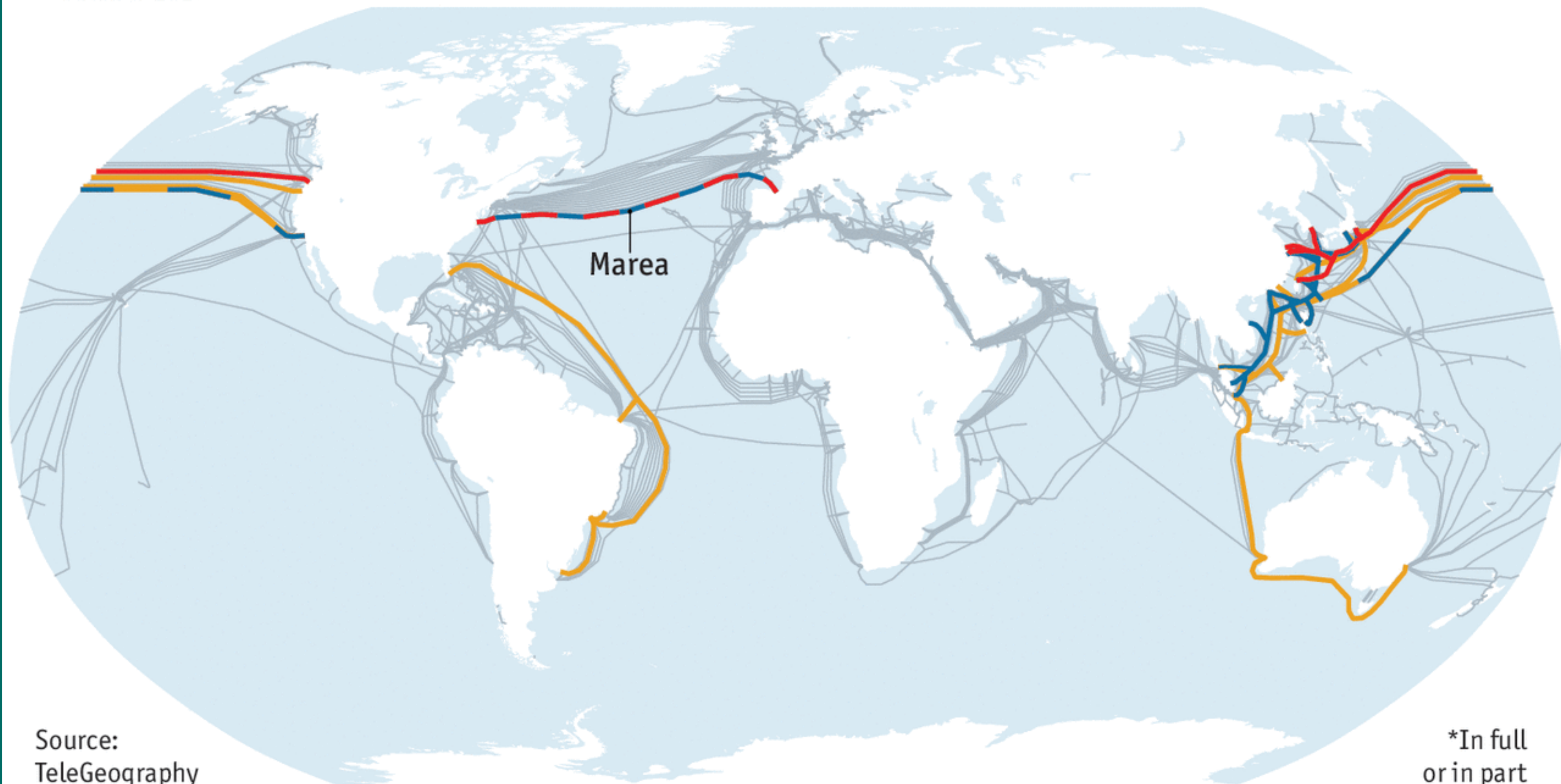
AWS

Azure

Facebook

Plumbing the depths

Active and planned submarine cable systems owned* by: — Facebook — Google — Microsoft — Other
October 2017



Source:
TeleGeography

*In full
or in part

IP ADDRESS
MAC ADDRESS
PORT NUMBERS

IPv4 ADDRESS

IP ADDRESS

- The logical address
- Must be unique in a LAN
- Private IP - to identify host within a LAN
- Public IP - to identify host on the Internet
- Can be assigned manually (static) or using DHCP (dynamic)
- IPv4: 192.168.1.1

MAC ADDRESS

- The physical address of Network Interface Card (NIC)
- Unique globally (no 2 NIC has the same MAC)
- To identify a host within a LAN
- Address Structure - 12:AB:CA:66:19:89

PORT NUMBERS

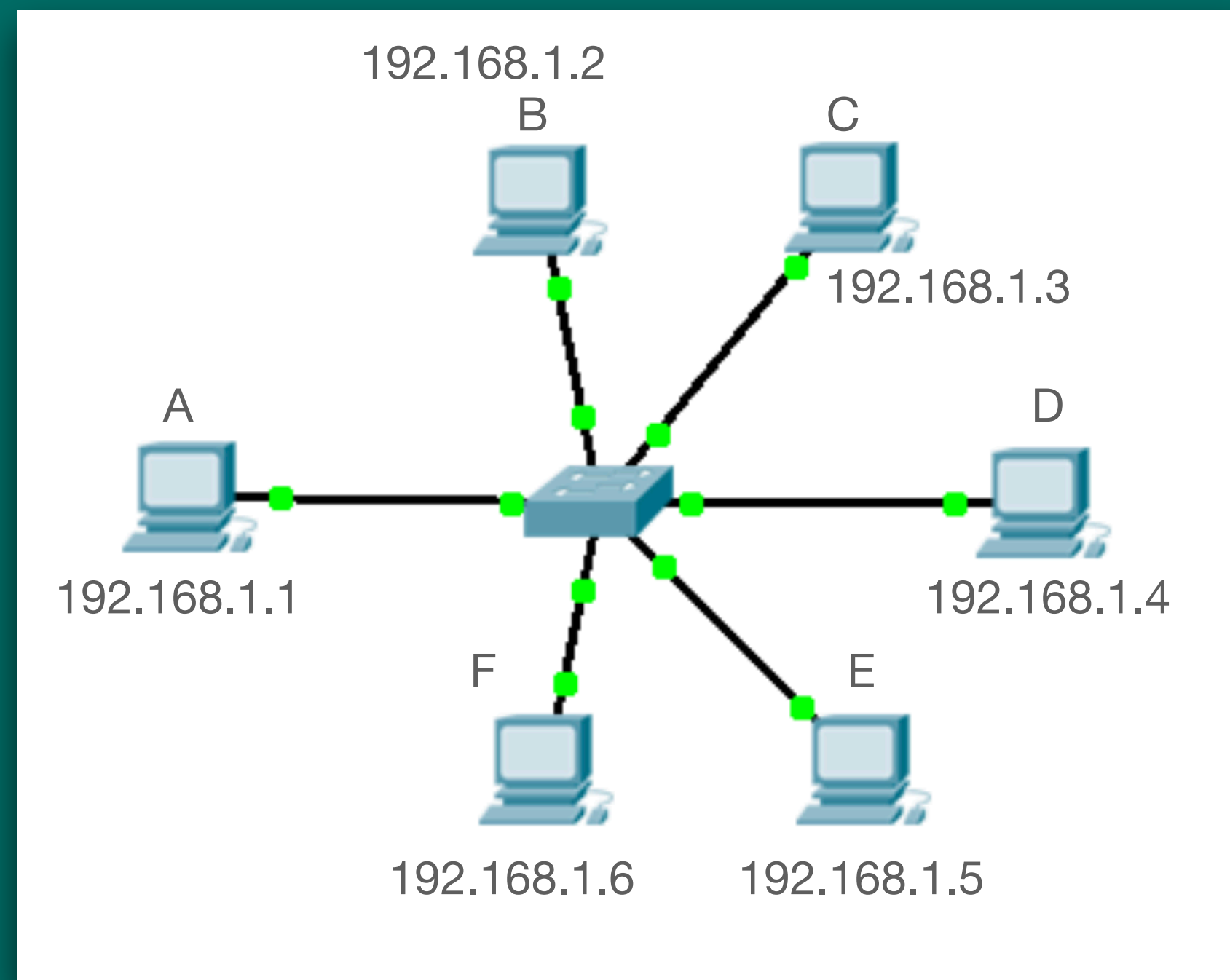
- Any numbers from 0 - 65535
- To identify the applications running in a host
- Well known ports: 0 - 1023
- Registered ports: 1024 - 49151
- Random ports: 49152 - 65535

IP ADDRESS

- The logical address to identify each host in a network
- IP address must be unique in a network
- Private IP - to identify host within a LAN
- Public IP - to identify host on the Internet
- Can be assigned manually (static) or using DHCP (dynamic)
- IPv4: 192.168.1.1

IP ADDRESS

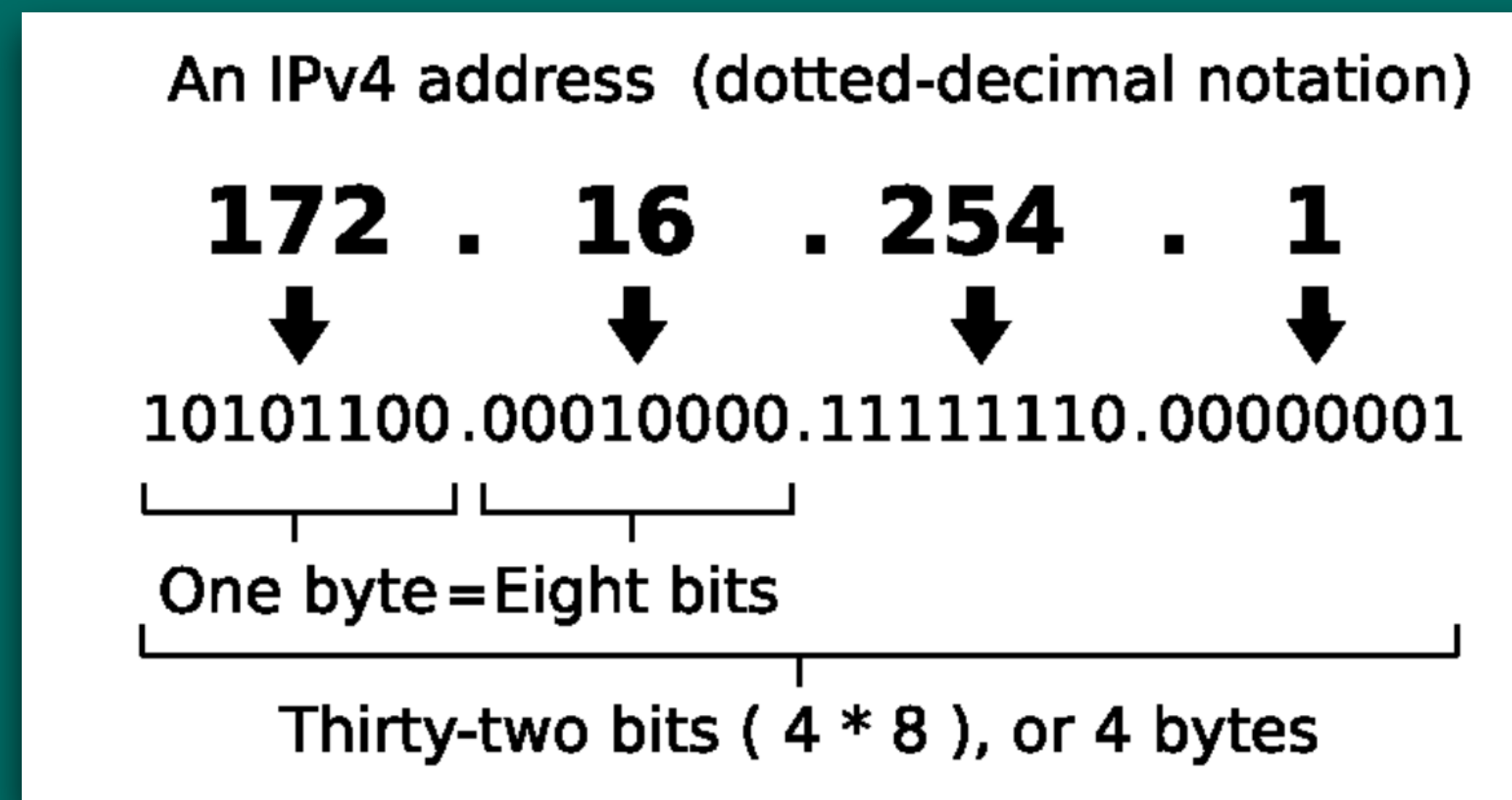
- The logical address to identify each host in a network
- IP address must be unique in a network



to 192.168.1.1
DATA

IPv4 ADDRESS

- There are two types of IP addresses: IPv4 and IPv6
- IPv4 is widely used these days, but we are migrating to IPv6 to support more devices.
- We only focus on IPv4 in datacom.
- IPv4 Address Structure:
 - each address is 32 bits long (4 bytes)
 - written as 4 numbers with 3 dots
 - each number is 8 bits.
 - total size: 4×8 bits = 32 bits
 - each number can range from 0—>255



IPv4 ADDRESS

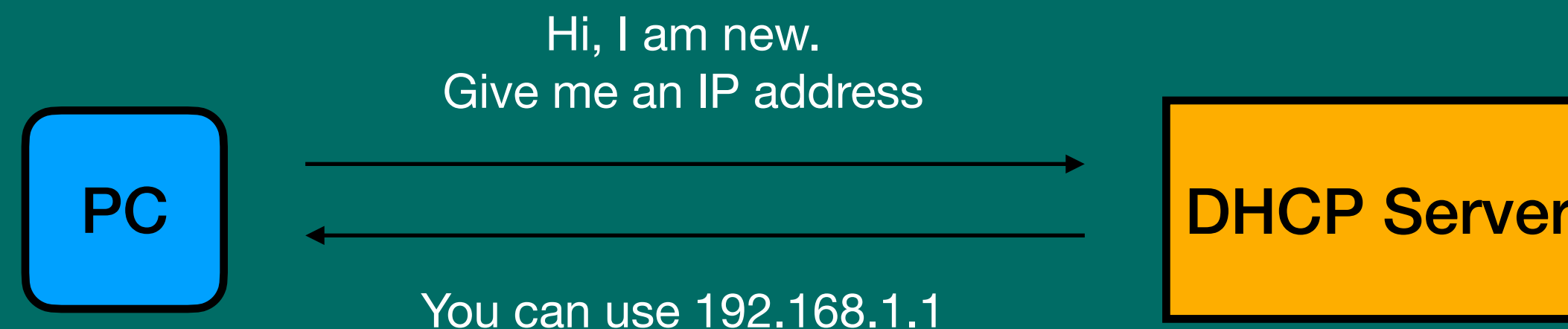
- IP addresses can be private or public
 - **Private IP** address is used within local area network
 - **Public IP** address is used to connect to the Internet
- **Private IP range:**

- Class A: 10.0.0.0 to 10.255.255.255
- Class B: 172.16.0.0 to 172.31.255.255
- Class C: 192.168.0.0 to 192.168.255.255

- Each device needs to have at least 1 private IP and 1 public IP to access the Internet
- **Q&A**
 - How to find your private IP on a PC?
 - How to find your public IP?
 - Can a device have two or more IP addresses?

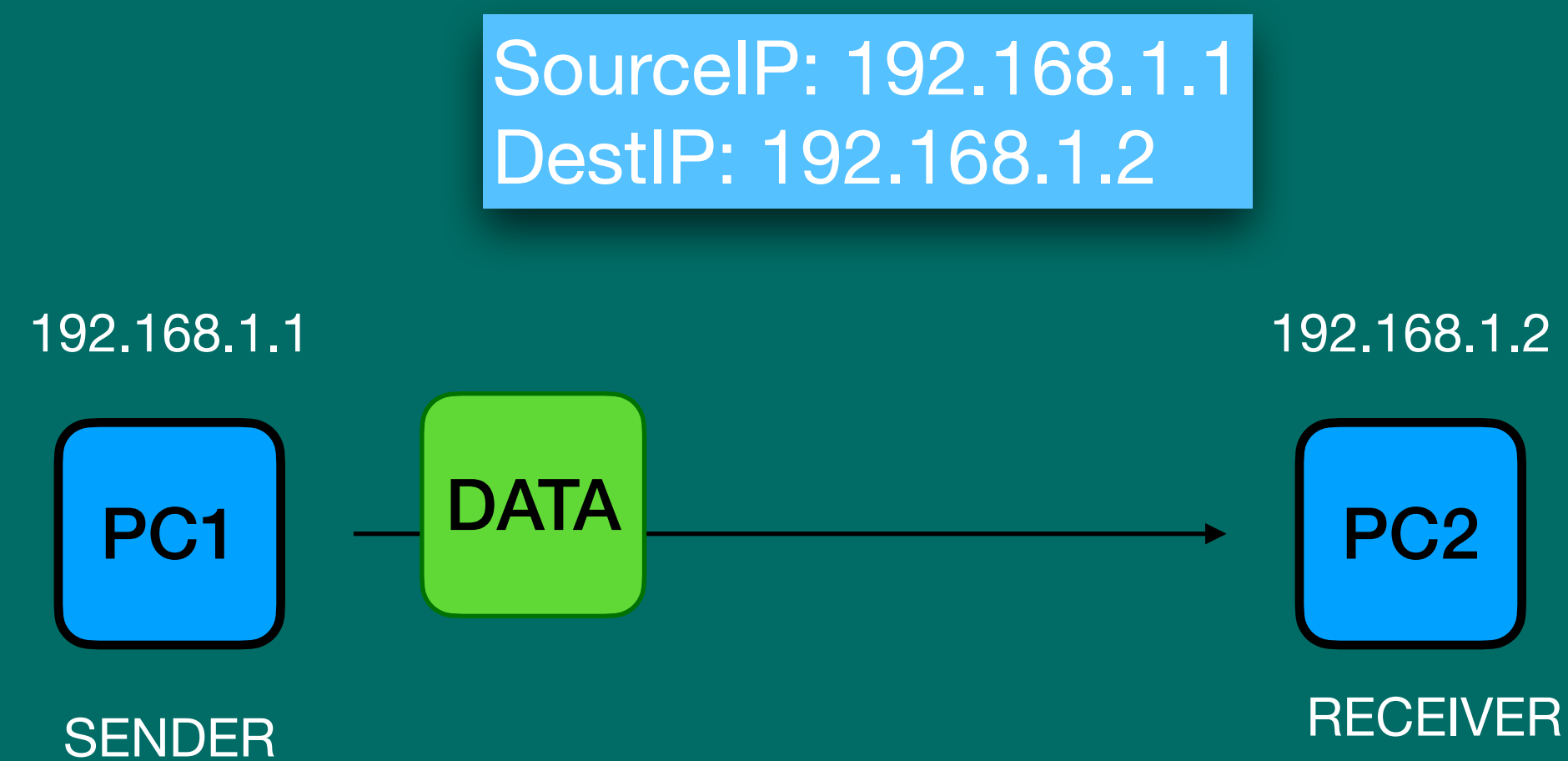
IPv4 ADDRESS

- IP addresses can be static or dynamic
 - Static IP address is configured manually by the user
 - Dynamic IP address is assigned automatically by a DHCP Server



IPv4 ADDRESS

- There are two types of IP addresses in data communications:
 - Source IP address is IP of the sender
 - Destination IP address is the IP of the receiver



MAC address

MAC ADDRESS

- The physical address of the Network Interface Card (NIC)
 - Unique globally (no 2 NIC has the same MAC)
 - To identify a host within a LAN
-
- MAC address has 6 bytes (or 48 bits).
 - The first 3 octet is to represent the manufacturer ID
 - The last 3 octet is the unique ID for each card

12 : AB : CA : 66 : 19 : 89

Manufacturer ID

Card ID

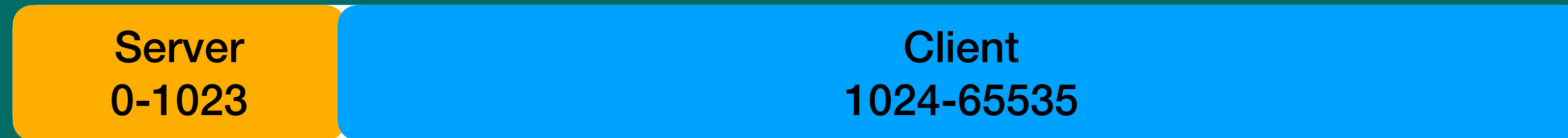
Port Numbers

Port Numbers

- Is used to identify applications
- Is a decimal number, ranging from 0-65535
- The size is 2 bytes

Port Numbers

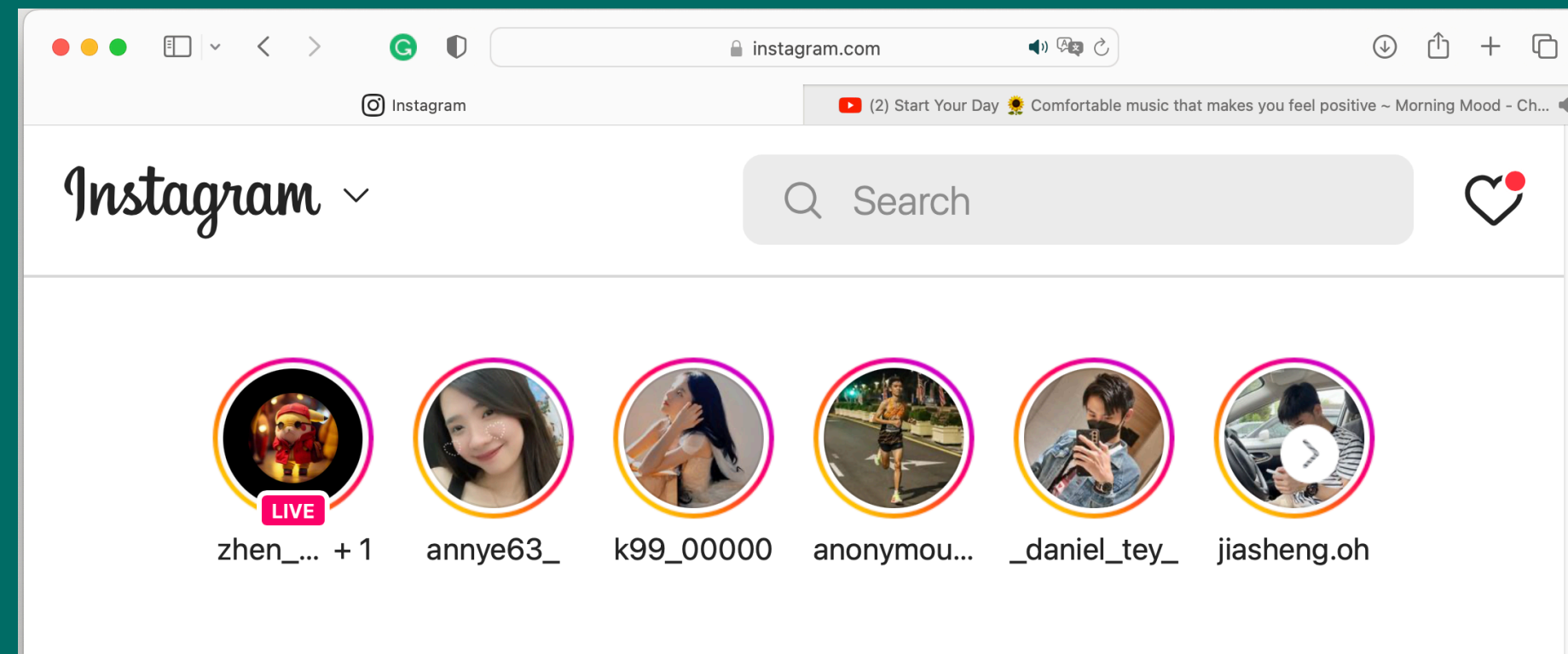
- There are client port and server port



- The server port is used for standard services like HTTP (80), TELNET (23), DNS(53).... Server port is assigned by IANA.
- The client port is used for by your devices (iPhone, iPad...) to run apps. Client port number is randomly assigned by the OS.

Port Numbers

- Client port vs server port



Client

Depending on browser
eg.182912

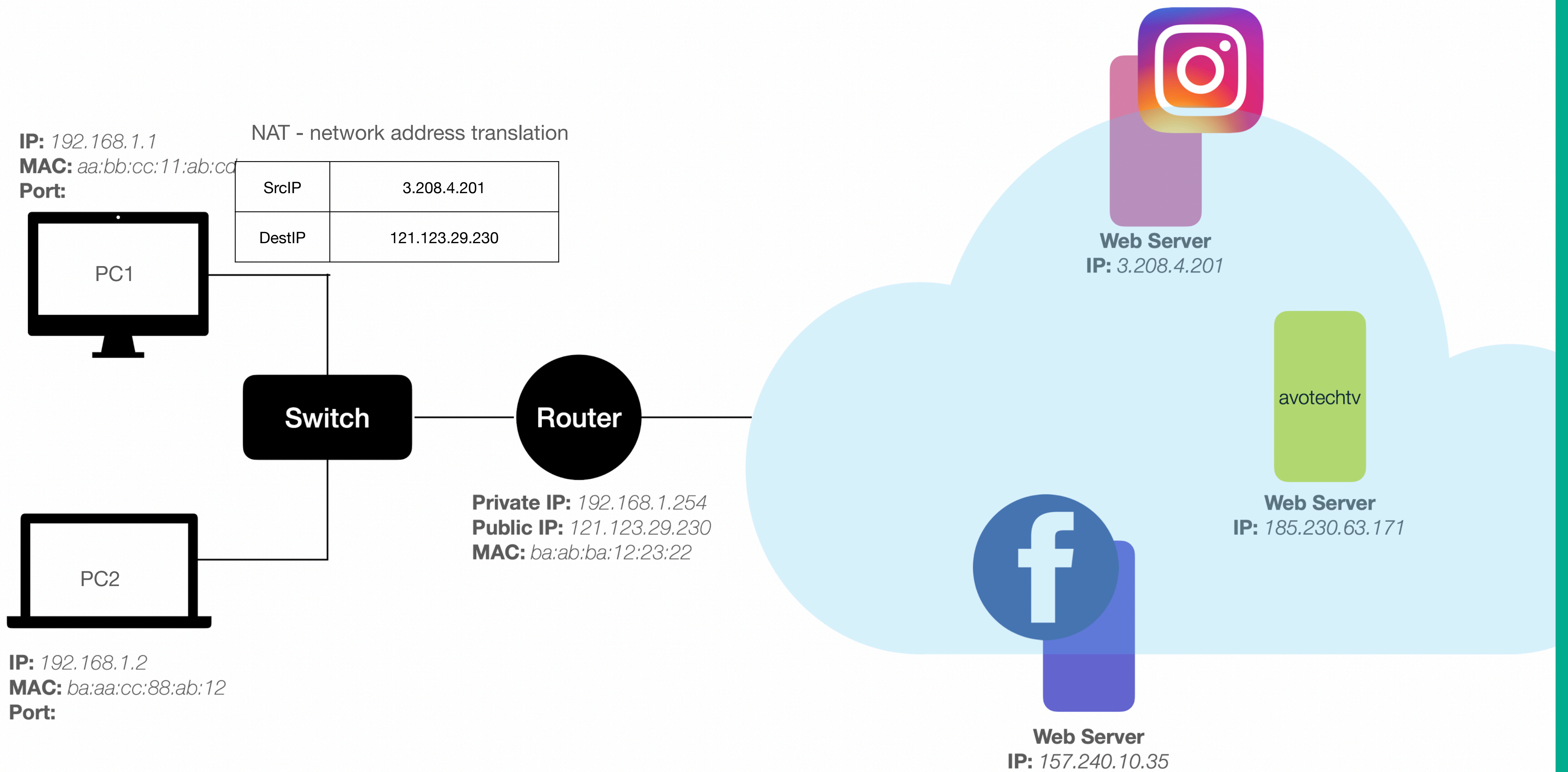


Instagram Web
(Port 80)

SERVER

HTTP server always run on port 80

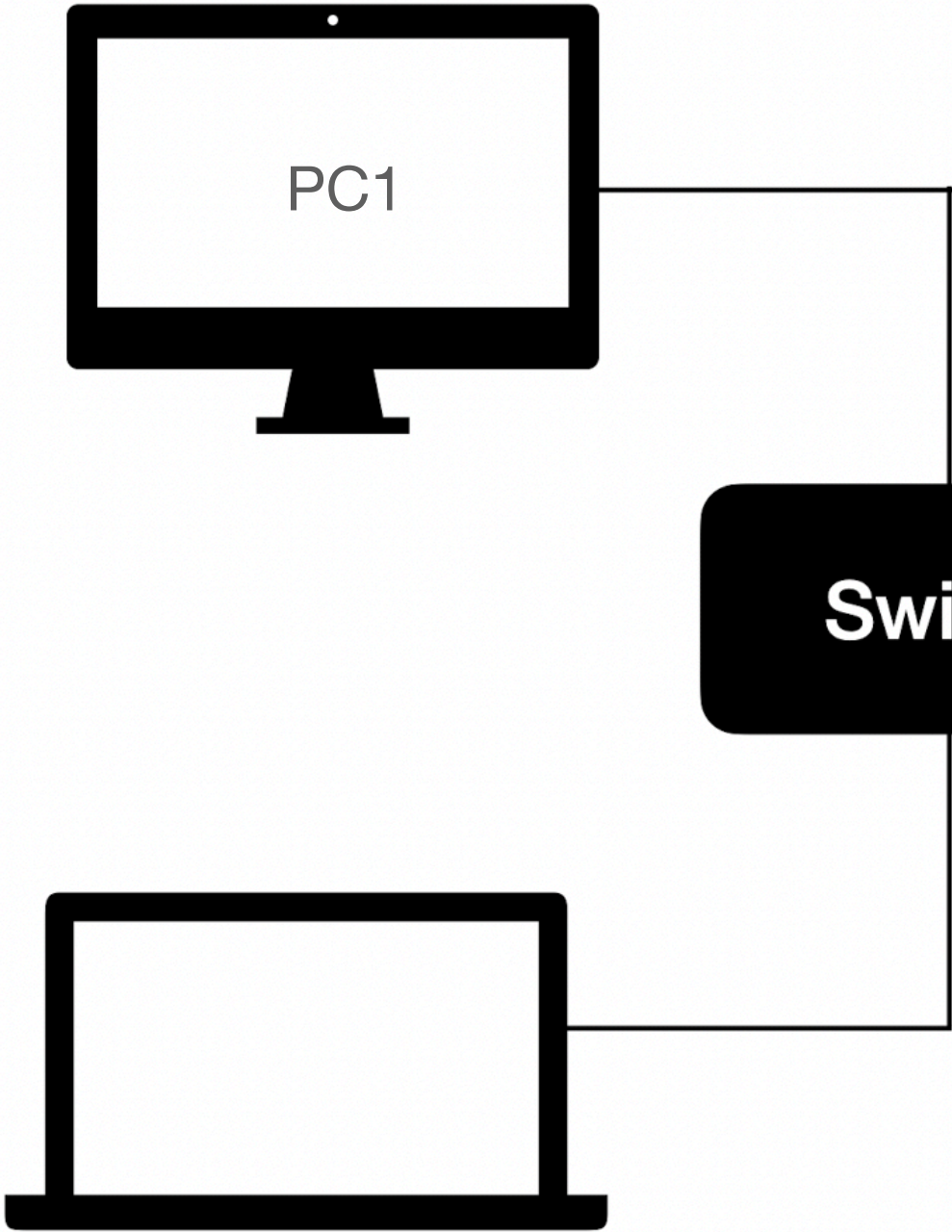
How IP + MAC + Port number
work together?



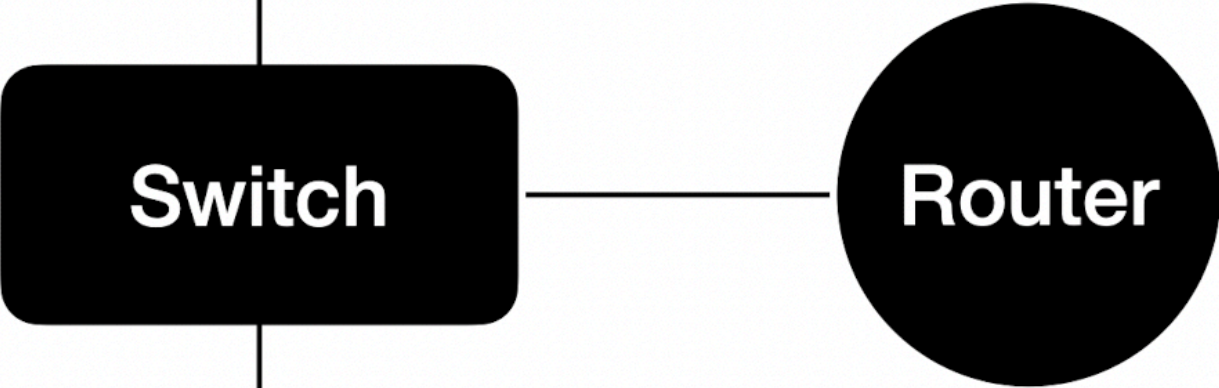
123 Facebook
1989 Instagram
8080 Avotechtv

IP: 192.168.1.1
MAC: aa:bb:cc:11:ab:cd
Port: 71234

SrcIP	192.168.1.1
DestIP	3.208.4.201
SrcPort	71234
DestPort	1989

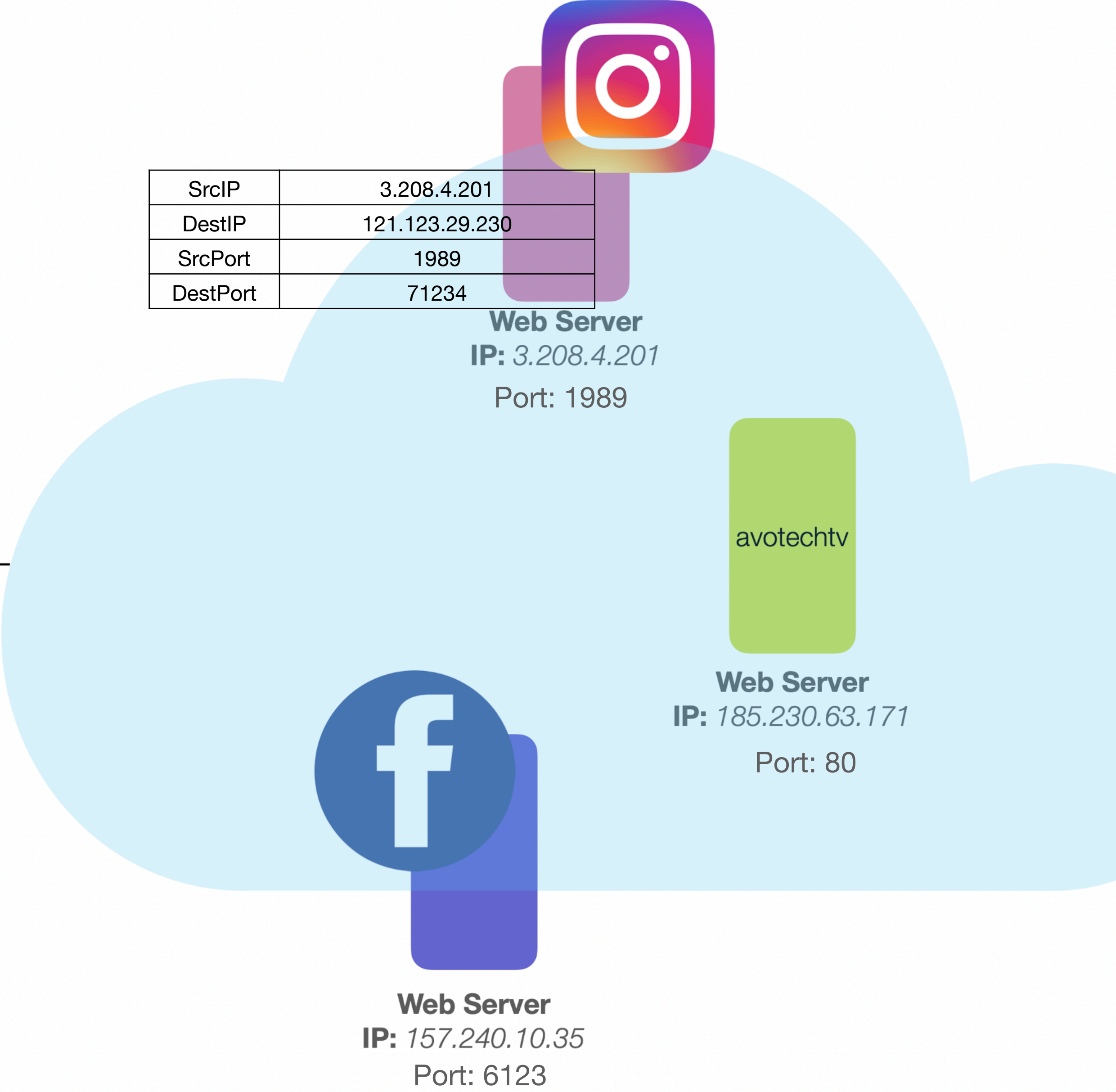


IP: 192.168.1.2
MAC: ba:aa:cc:88:ab:12
Port:



Private IP: 192.168.1.254
Public IP: 121.123.29.230
MAC: ba:ab:ba:12:23:22

SrcIP	3.208.4.201
DestIP	121.123.29.230
SrcPort	1989
DestPort	71234



Build your own LAN (Bob the Builder on THM)